

VENDOR OUTSTANDING BALANCE DASHBOARD

by Ishina Labit



Recreation and Data Disclaimer

This project is a recreation of a project completed during an internship. Due to the confidential nature of the original financial data and systems, synthetic data and a local SQL Server environment were used to recreate the general data-processing and reporting workflow.

The dataset differs substantially from the original organizational data in structure, values, volume, and appearance. This recreation is intended solely to demonstrate the technical approach, process, and reporting objectives of the original project and does not contain or reproduce any proprietary company data.

VENDOR OUTSTANDING BALANCE

PROJECT OBJECTIVE

The objective of the project is to improve the visibility and accuracy of vendor outstanding balance reporting in support of the company's finance and accounting departments and their operations. The solution enables users to analyze outstanding balances by Buyer Group IDs and quickly identify high-balance accounts.

DATA SOURCE

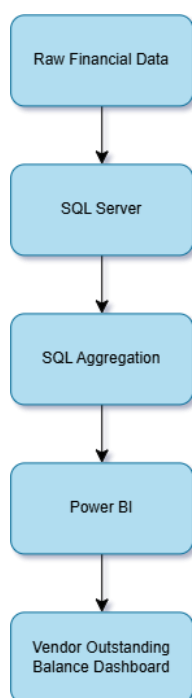


Figure 1: Data processing architecture

The data source for this project is a simplified version of the dataset I originally worked with. This project includes a table with 10,000 rows of synthetic financial transactional data containing information about vendors, Buyer Group IDs, invoices, payments, and transaction dates.

Below are the columns and their respective descriptions:

TRANSACTION TABLE VIEW

TransactionID	Key identifying a specific financial transaction
VendorID	Key that helps identify the vendors in the transaction
Vendor Name	Name of the vendor filed in the system

VENDOR OUTSTANDING BALANCE

BuyerGroupID	Identifies buyers (customers) who are treated as a single purchasing group
InvoiceID	A specific invoice billed to a buyer from the vendor
Invoice Date	Date when the transaction takes place
Due Date	Date when the buyer is expected to settle the bill before incurring penalties
Invoice Amount	A monetary transactional amount expected to be paid by the buyer to the vendor
Paid Amount	How much a buyer has paid per transaction
Outstanding Balance	Calculated by [Invoice Amount] – [Paid Amount]
PaymentStatus	Is classified as one of: Outstanding, Partially Paid, or Paid

SQLQuery5....riss (80) x SQLQuery3....kriss (68)*

```

1  SELECT TOP (1000) [TransactionID]
2      , [Buyer Group ID]
3      , [Vendor ID]
4      , [Vendor Name]
5      , [Invoice ID]
6      , [Invoice Date]
7      , [Due Date]
8      , [Invoice Amount]
9      , [Paid Amount]
10     , [Outstanding Balance]
11     , [Payment Status]
12 FROM [PersonalProjects].[dbo].[VendorTransactions]
13

```

161 % 12 0 ↑ ↓

	TransactionID	Buyer Group ID	Vendor ID	Vendor Name	Invoice ID	Invoice Date	Due Date	Invoice Amount	Paid Amount	Outstanding Balance	Payment Status
1	1	BG-101	V003	CoreTech Solutions	INV-100001	2026-06-29	2025-11-30	37789.00	12092.48	25696.52	Partially Paid
2	2	BG-105	V020	IronWorks Engineering	INV-100002	2025-11-24	2026-02-01	70719.00	70719.00	0.00	Paid
3	3	BG-101	V001	Apex Industrial Supplies	INV-100003	2025-09-13	2026-01-29	75424.00	0.00	75424.00	Outstanding
4	4	BG-103	V015	United Maintenance	INV-100004	2026-08-11	2026-06-27	23847.00	23847.00	0.00	Paid
5	5	BG-102	V017	Westfield Distribution	INV-100005	2026-06-07	2026-06-21	74942.00	19484.92	55457.08	Partially Paid
6	6	BG-106	V008	Horizon IT Services	INV-100006	2026-03-20	2026-01-13	92524.00	0.00	92524.00	Outstanding
7	7	BG-107	V010	NorthStar Consulting	INV-100007	2025-11-12	2026-02-05	89100.00	26730.00	62370.00	Partially Paid
8	8	BG-102	V002	BlueLine Logistics	INV-100008	2025-12-29	2026-04-29	82250.00	0.00	82250.00	Outstanding
9	9	BG-104	V005	East Coast Packaging	INV-100009	2026-01-30	2026-08-07	16270.00	10250.10	6019.90	Partially Paid
10	10	BG-103	V015	United Maintenance	INV-100010	2025-10-24	2025-09-20	77663.00	0.00	77663.00	Outstanding
11	11	BG-103	V004	Delta Office Systems	INV-100011	2025-10-20	2026-05-25	76985.00	0.00	76985.00	Outstanding
12	12	BG-101	V003	CoreTech Solutions	INV-100012	2026-02-20	2026-08-07	59813.00	0.00	59813.00	Outstanding

Figure 2: SSMS view of the Transactions table

VENDOR OUTSTANDING BALANCE

SQL DATA PREPARATION

The SQL Server was set up in a local environment where the dataset, VendorTransaction, was created. The data contained in VendorTransactions is entirely synthetic and was generated specifically for this recreation.

The VendorTransactions table was used as the transaction-level dataset. SQL was then used to aggregate the transaction data by Buyer Group ID and Vendor ID. A SQL view, vw_OutstandingBalance, was created to aggregate the transaction-level data into a reporting-ready format, allowing Power BI to efficiently access the relevant financial information for analysis and visualization.

The following SQL syntax was used to create the view:

```
CREATE VIEW vw_OutstandingBalances AS
SELECT
    [Buyer Group ID],
    [Vendor ID],
    [Vendor Name],
    COUNT(DISTINCT [Invoice ID]) AS InvoiceCount,
    SUM([Invoice Amount]) AS TotalInvoiceAmount,
    SUM([Paid Amount]) AS TotalPaidAmount,
    SUM([Outstanding Balance]) AS OutstandingBalance
FROM VendorTransactions
GROUP BY
    [Buyer Group ID],
    [Vendor ID],
    [Vendor Name];
```

POWER BI CONNECTION

After creating the OutstandingBalances view, Power BI is then connected to the local SQL Server using the SQL Server connector. Instead of importing the raw transaction table, the view was used as the primary data source for the Power BI dashboard, reducing the amount of data required for the visualizations.

VENDOR OUTSTANDING BALANCE

DASHBOARD DESIGN

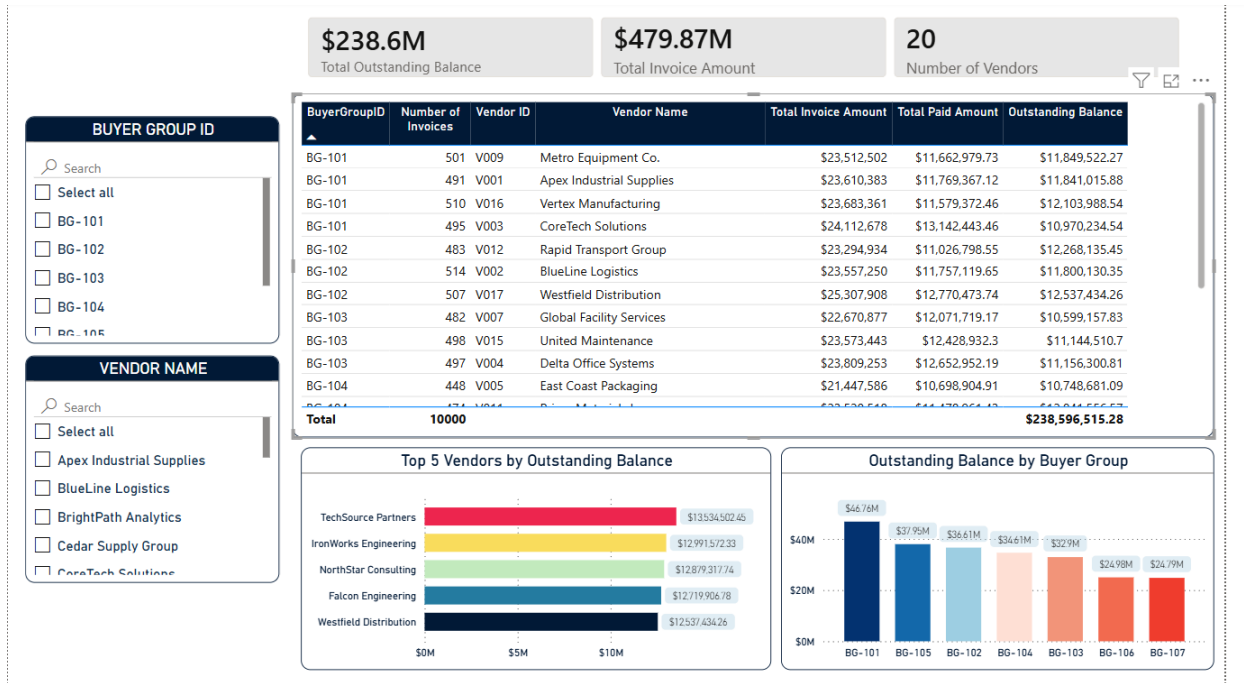


Figure 3: Vendor Outstanding Balance Dashboard

The dashboard design includes a table visualization of the OutstandingBalance SQL view, a horizontal bar chart for the Top 5 Vendors by Outstanding Balance, and a bar chart highlighting the Outstanding Balance per Buyer Group. Above the charts are cards showing the Total Outstanding Balance, Total Invoice Amount, and the Total Number of Vendors. For easier filtering, slicers for Buyer Group IDs and Vendor Names were included.

DATA VALIDATION

To validate the accuracy of the dashboard, aggregate outstanding balances were compared between the VendorTransactions table and Power BI.

	SQL Result	Power BI Result
Total Outstanding Balance	\$238,596,515.28	\$238,596,515.28
Number of Vendors	20	20

VENDOR OUTSTANDING BALANCE

Buyer Group BG-102	\$36,605,700.06	\$36,605,700.06
-----------------------	-----------------	-----------------

Below is the SQL syntax used to validate the information:

-- Total Outstanding Validation

```
SELECT SUM([Outstanding Balance])  
FROM VendorTransactions
```

-- Total Number of Vendors

```
SELECT  
    COUNT(DISTINCT([Vendor ID])) AS Total Number of Vendors  
--FROM VendorTransactions
```

-- Outstanding Balance for BG-102

```
SELECT  
    [Buyer Group ID],  
    SUM([Outstanding Balance]) AS [Total Outstanding Bal]  
FROM VendorTransactions  
WHERE [Buyer Group ID] = 'BG-102'  
GROUP BY [Buyer Group ID];
```

OUTCOME

The resulting Power BI dashboard provides a centralized view of vendor outstanding balances per Buyer Group ID. By utilizing SQL data transformation and aggregation, the dashboard solution improves reporting efficiency and enables users to quickly identify high-balance vendor accounts.